DIFFERENTIALS, DUAL ISOGENIES, AND THE TATE MODULE: SOLUTIONS

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Here are three computations by hand, on the invariant differential, the dual isogeny and the Tate module. Each compares the multiplicative group $\mathbb{G}_{\mathrm{m}}$ with the elliptic curve

$$E\colon y^2 = x^3 - x, \qquad \omega_E \coloneqq \frac{\mathrm{d}x}{2y}.$$

Throughout, k is a field with $\operatorname{char} k \neq 2$, $\overline{k}$ an algebraic closure and $k^{\mathrm{sep}} \subseteq \overline{k}$ the separable closure. Here $\mathbb{G}_{\mathrm{m}} = \mathbb{P}^1_k \smallsetminus \{0, \infty\}$ has coordinate t and invariant differential $\omega_{\mathbb{G}_{\mathrm{m}}} \coloneqq \mathrm{d}t/t$. Each problem is preceded by the definitions and results it uses.

Reading. [Sil09, §§III.4–III.7], and [Sil09, §III.8] for the Weil pairing of Proposition 3.1.

A formula. We use the following duplication formula on E, derived in Problem 1.1(5): if $x \in k(E)$ is the x-coordinate and $[2]\colon E \to E$ is multiplication by 2, then

$$[2]^* x \coloneqq x \circ [2] = \frac{(x^2+1)^2}{4(x^3 - x)}. \tag{0.1}$$

1. Invariant differentials

Problem 1.1 (Poles and pullbacks)**.**

(1) Rewrite $\omega_{\mathbb{G}_{\mathrm{m}}}$ in the coordinate $s \coloneqq 1/t$ near ∞, and compute $\operatorname{div}(\omega_{\mathbb{G}_{\mathrm{m}}})$ on $\mathbb{P}^1_k$. Compute $[m]^*\omega_{\mathbb{G}_{\mathrm{m}}}$ for $[m](t) = t^m$, $m \in \mathbb{Z}$.

Solution. $\boldsymbol{\omega_{\mathbb{G}_{\mathrm{m}}} = -\,\mathrm{d}s/s}$**,** $\boldsymbol{\operatorname{div}(\omega_{\mathbb{G}_{\mathrm{m}}}) = -(0) - (\infty)}$ **and** $\boldsymbol{[m]^*\omega_{\mathbb{G}_{\mathrm{m}}} = m\,\omega_{\mathbb{G}_{\mathrm{m}}}}$**.** $t = 1/s$ gives $\mathrm{d}t = -\,\mathrm{d}s/s^2$, so $\omega_{\mathbb{G}_{\mathrm{m}}} = -\,\mathrm{d}s/s$. Hence $\omega_{\mathbb{G}_{\mathrm{m}}}$ has simple poles at 0 and ∞, with residues 1 and -1, and no other zeros or poles: $\operatorname{div}(\omega_{\mathbb{G}_{\mathrm{m}}}) = -(0) - (\infty)$. Also $[m]^*\omega_{\mathbb{G}_{\mathrm{m}}} = \mathrm{d}(t^m)/t^m = m\,\omega_{\mathbb{G}_{\mathrm{m}}}$.

(2) Show that $\omega_E = \mathrm{d}y/(3x^2 - 1)$, and that y and $3x^2 - 1$ have no common zero on E. Deduce that ω_E has no zeros or poles on $E \smallsetminus \{O\}$.

Solution. Differentiating $y^2 = x^3 - x$ gives $2y\,\mathrm{d}y = (3x^2 - 1)\,\mathrm{d}x$. If $y = 0$, then $x \in \{0, \pm 1\}$ and $3x^2 - 1 \in \{-1, 2\}$, which is non-zero. On the open set $y \neq 0$ the differential $\mathrm{d}x$ generates $\Omega_{E/k}$, and $\omega_E = \mathrm{d}x/(2y)$ is a unit times it. On $3x^2 - 1 \neq 0$, likewise with $\mathrm{d}y$ and $\mathrm{d}y/(3x^2 - 1)$. These open sets cover $E \smallsetminus \{O\}$.

(3) Using $\operatorname{ord}_O x = -2$ and $\operatorname{ord}_O y = -3$, show that $\operatorname{ord}_O \omega_E = 0$.

Solution. With a uniformizer w at O, write $x = w^{-2}u$ with $u(O) \neq 0$. Then

$$\mathrm{d}x = (-2w^{-3}u + w^{-2}u')\,\mathrm{d}w$$

has order -3, because $2 \neq 0$ in k. So $\operatorname{ord}_O \omega_E = -3 - (-3) = 0$.

(4) Compute $[-1]^*\omega_E$ from $[-1](x, y) = (x, -y)$, and compare with $[-1]^*\omega_{\mathbb{G}_{\mathrm{m}}}$.

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Solution. **$[-1]^*\omega_E = -\omega_E$, and $[-1]^*\omega_{\mathbb{G}_m} = -\omega_{\mathbb{G}_m}$.** $[-1]^*\omega_E = \mathrm{d}x/(-2y) = -\omega_E$. On $\mathbb{G}_m$, $[-1](t) = 1/t$ and $[-1]^*\omega_{\mathbb{G}_m} = -\omega_{\mathbb{G}_m}$ as well.

(5) We derive (0.1) by hand. Here $x, y \in k(E)$ are functions. For a point $P \in E(\overline{k})$ write $x_P \coloneqq x(P)$ and $y_P \coloneqq y(P)$ for its coordinates.

(a) Let $P \in E(\overline{k})$ with $y_P \neq 0$. Show that the tangent line to E at P has slope $\lambda_P = (3x_P^2 - 1)/(2y_P)$, and use that its third point of intersection with E is $-[2]P$ to show that $x([2]P) = \lambda_P^2 - 2x_P$.

Solution. In the affine plane with coordinates x, y, differentiating $y^2 = x^3 - x$ at P gives $2y_P\,\mathrm{d}y = (3x_P^2 - 1)\,\mathrm{d}x$, so the tangent line at P is

$$y = \lambda_P x + \nu_P, \qquad \lambda_P = \frac{3x_P^2 - 1}{2y_P}, \qquad \nu_P = y_P - \lambda_P x_P.$$

Substituting into $y^2 = x^3 - x$ gives

$$x^3 - \lambda_P^2 x^2 - (1 + 2\lambda_P\nu_P)x - \nu_P^2 = 0,$$

whose roots are x_P, twice because the line is tangent, and $x(-[2]P)$. Comparing coefficients of x^2 gives $x(-[2]P) = \lambda_P^2 - 2x_P$, and $x(-[2]P) = x([2]P)$ since $[-1](x, y) = (x, -y)$.

(b) Let $\lambda \coloneqq (3x^2 - 1)/(2y) \in k(E)$, so that $\lambda(P) = \lambda_P$. Explain why (5)(a) gives the identity $[2]^*x = \lambda^2 - 2x$ of functions in $k(E)$, and simplify it using $y^2 = x^3 - x$.

Solution. **$[2]^*x = (x^2 + 1)^2/(4(x^3 - x))$.** By definition of the pullback, $([2]^*x)(P) = x([2]P)$, and $(\lambda^2 - 2x)(P) = \lambda_P^2 - 2x_P$. So by (5)(a) the function $[2]^*x - (\lambda^2 - 2x) \in k(E)$ vanishes at every $P \in E(\overline{k})$ with $y_P \neq 0$, that is, at every point of $E(\overline{k})$ outside the finite set $E[2](\overline{k})$. A non-zero rational function has only finitely many zeros, so $[2]^*x = \lambda^2 - 2x$ in $k(E)$. Finally, in $k(E)$ we have $4y^2 = 4(x^3 - x)$, so

$$[2]^*x = \lambda^2 - 2x = \frac{(3x^2 - 1)^2 - 8x(x^3 - x)}{4(x^3 - x)} = \frac{x^4 + 2x^2 + 1}{4(x^3 - x)} = \frac{(x^2 + 1)^2}{4(x^3 - x)}.$$

Remark. This is the precise meaning of the shorthand "$x([2]P) = \lambda^2 - 2x$ for $P = (x, y)$".

(6) Solve this part in `Magma`, working in the function field $\mathbb{Q}(E) = \mathbb{Q}(x)[y]/(y^2 - x^3 + x)$. Compute $[2]^*x$ and $[2]^*y$ from the function λ of (5), check (0.1), and show that

$$[2]^*x - e = \frac{g_e(x)^2}{4(x^3 - x)}, \qquad g_0 = x^2 + 1, \quad g_1 = x^2 - 2x - 1, \quad g_{-1} = x^2 + 2x - 1,$$

for $e \in \{0, 1, -1\}$. Show that $[2]^*y = \pm(x^2+1)(x^4-6x^2+1)/(8y^3)$ and $[2]^*\omega_E = \pm 2\omega_E$. Which sign is the correct one?

Solution. **Both signs are $+$.** The identities have coefficients in $\mathbb{Z}[1/2]$, so it suffices to check them over $\mathbb{Q}$. By (5), $[2]^*x = \lambda^2 - 2x$ with $\lambda = (3x^2 - 1)/(2y)$, and the same argument applied to the y-coordinate of the third point of intersection gives $[2]^*y = -\big(y + \lambda([2]^*x - x)\big)$. Since $\omega_E = \mathrm{d}x/(2y)$, we have

$$\frac{[2]^*\omega_E}{\omega_E} = \frac{\mathrm{d}([2]^*x)/\,\mathrm{d}x}{2\,[2]^*y} \cdot 2y.$$

```
Kx<x> := FunctionField(Rationals());
R<t>  := PolynomialRing(Kx);
F<y>  := ext<Kx | t^2 - (x^3 - x)>;                    // Q(E)
lam := (3*x^2 - 1)/(2*y);                              // tangent slope
```

```
x2 := lam^2 - 2*x;  y2 := -(y + lam*(x2 - x));          // [2]^*x, [2]^*y
x2 eq (x^2 + 1)^2/(4*(x^3 - x));                        // true
[ Factorization(Numerator(Kx!(x2 - e))) : e in [0, 1, -1] ];
// [ [ <x^2 + 1, 2> ], [ <x^2 - 2*x - 1, 2> ], [ <x^2 + 2*x - 1, 2> ] ]
y2 eq (x^2 + 1)*(x^4 - 6*x^2 + 1)/(8*y^3);              // true
(F!Derivative(Kx!x2))/(2*y2) * 2*y;                     // 2
```

The denominator of $[2]^*x - e$ is that of $[2]^*x$, namely $4(x^3 - x)$, and the factorizations show that its numerator is $g_e(x)^2$. So $[2]^*y = (x^2+1)(x^4 - 6x^2 + 1)/(8y^3)$ and $[2]^*\omega_E = 2\omega_E$, as [Sil09, Corollary III.5.3] predicts.

Remark. Alternatively, `Magma`'s elliptic curve functionality provides [2] as an isogeny, and pulls back both functions and differentials along it, so no group law is typed in.

```
E := EllipticCurve([-1,0]);                             // y^2 = x^3 - x
m2 := MultiplicationByMMap(E, 2);                       // [2] : E -> E
KE<x,y> := FunctionField(E);                            // Q(E)
x2 := Pullback(m2, x);  y2 := Pullback(m2, y);          // [2]^*x, [2]^*y
x2 eq (x^2 + 1)^2/(4*(x^3 - x));                        // true
g := [x^2 + 1, x^2 - 2*x - 1, x^2 + 2*x - 1];           // g_0, g_1, g_{-1}
[ x2 - [0, 1, -1][i] eq g[i]^2/(4*(x^3 - x)) : i in [1..3] ];
// [ true, true, true ]
y2 eq (x^2 + 1)*(x^4 - 6*x^2 + 1)/(8*y^3);              // true
omega := Differential(x)/(2*y);                         // omega_E
Pullback(m2, omega)/omega;                              // 2
```

2. Dual isogenies

Consider the elliptic curves

$$E_1\colon y_1^2 = x_1^3 - x_1, \qquad E_2\colon y_2^2 = x_2^3 + 4x_2, \qquad E_3\colon y_3^2 = x_3^3 - 16x_3,$$

so that $E_1 = E$, and write $\omega_i := \mathrm{d}x_i/(2y_i)$ for the invariant differential of E_i. Consider the maps

$$\begin{aligned}
\varphi\colon E_1 \to E_2, &\quad (x_1, y_1) \mapsto \left(\frac{x_1^2 - 1}{x_1},\ \frac{y_1(x_1^2+1)}{x_1^2}\right),\\
\psi\colon E_2 \to E_3, &\quad (x_2, y_2) \mapsto \left(\frac{y_2^2}{x_2^2},\ \frac{y_2(x_2^2-4)}{x_2^2}\right),\\
\iota\colon E_3 \to E_1, &\quad (x_3, y_3) \mapsto \left(\frac{x_3}{4},\ \frac{y_3}{8}\right).
\end{aligned}$$

The maps φ and ψ are special cases of [Sil09, Example III.4.5], which treats $y^2 = x^3 + ax^2 + bx$ in general, so they are isogenies. Also ι is an isomorphism. That example computes the kernels and the dual, but not the constants c_α of Definition 2.1: that is part (2) of Problem 2.2.

Definition 2.1 (The constant c_α)**.** For an isogeny $\alpha\colon E_i \to E_j$, define c_α by $\alpha^*\omega_j = c_\alpha\,\omega_i$.

Problem 2.2 (A 2-isogeny and its dual)**.**
(1) Check that φ lands on E_2. Show that $\ker\varphi = \{O_1, (0,0)\}$, that $\varphi(\pm 1, 0) = (0,0)$, and that $\deg\varphi = 2$.

SOLUTION. Since $y_1^2 = x_1(x_1^2-1)$, the functions $\varphi^* x_2 = (x_1^2-1)/x_1$ and $\varphi^* y_2 = y_1(x_1^2+1)/x_1^2$ satisfy

$$(\varphi^* y_2)^2 = \frac{x_1(x_1^2-1)(x_1^2+1)^2}{x_1^4}, \qquad (\varphi^* x_2)^3 + 4\varphi^* x_2 = \frac{x_1^2-1}{x_1}\cdot\frac{(x_1^2+1)^2}{x_1^2},$$

which agree. The function $\varphi^* x_2 = x_1 - 1/x_1$ has poles only at $(0,0)$ and at O_1, so $\ker\varphi = \{O_1, (0,0)\}$. And at $(\pm 1, 0)$ both coordinates of φ vanish. Next,

$$y_1 = \frac{x_1^2\,\varphi^* y_2}{x_1^2+1}, \qquad x_1^2 - (\varphi^* x_2)x_1 - 1 = 0,$$

so $k(E_1) = \varphi^* k(E_2)(x_1)$ has degree at most 2 over $\varphi^* k(E_2)$. Since $\#\ker\varphi = \deg_s\varphi \leqslant \deg\varphi$, we get $\deg\varphi = 2$.

(2) Compute c_φ.

SOLUTION. $\boldsymbol{c_\varphi = 1.}$

$$\varphi^*\,\mathrm{d}x_2 = \left(1 + \frac{1}{x_1^2}\right)\mathrm{d}x_1 = \frac{(x_1^2+1)\,\mathrm{d}x_1}{x_1^2}, \qquad 2\varphi^* y_2 = \frac{2y_1(x_1^2+1)}{x_1^2},$$

so $\varphi^*\omega_2 = \omega_1$ and $c_\varphi = 1$.

Remark. This is why the second coordinate of φ has the sign it does: the opposite sign would give $c_\varphi = -1$.

(3) Check that ψ lands on E_3 and that ι lands on E_1, and compute c_ψ and c_ι.

SOLUTION. $\boldsymbol{c_\psi = 1}$ **and** $\boldsymbol{c_\iota = 2.}$ φ and ψ are the same construction for $y^2 = x^3 + bx$, with $b = -1$ and $b = 4$ respectively, and target $y^2 = x^3 - 4bx$. The computation of (1) applies verbatim, and that of (2) gives $c_\psi = 1$. Also $(y_3/8)^2 = (x_3^3 - 16x_3)/64 = (x_3/4)^3 - x_3/4$, so ι lands on E_1, and

$$\iota^*\omega_1 = \frac{\mathrm{d}x_3/4}{2y_3/8} = 2\,\omega_3,$$

so $c_\iota = 2$.

(4) Show that $(\iota\circ\psi\circ\varphi)^* x_1 = [2]^* x_1$, so that $\iota\circ\psi\circ\varphi = \pm[2]$. Use (2) and (3) to decide the sign, and conclude that $\hat\varphi = \iota\circ\psi$, that $c_\varphi c_{\hat\varphi} = \deg\varphi$, and that $\ker\hat\varphi = \varphi(E_1[2](\bar k))$.

SOLUTION. **The sign is +:** $\boldsymbol{\iota\circ\psi\circ\varphi = [2]}$**, and** $\boldsymbol{\hat\varphi = \iota\circ\psi.}$ We have $\psi^* x_3 = y_2^2/x_2^2 = x_2 + 4/x_2$ and $\iota^* x_1 = x_3/4$, so

$$(\iota\circ\psi\circ\varphi)^* x_1 = \frac{1}{4}\left(\frac{x_1^2-1}{x_1} + \frac{4x_1}{x_1^2-1}\right) = \frac{(x_1^2+1)^2}{4(x_1^3-x_1)},$$

which is $[2]^* x_1$ by (0.1). Two isogenies with the same pullback of x_1 agree up to sign. Now $c_{\iota\circ\psi\circ\varphi} = 2\cdot 1\cdot 1 = 2 = c_{[2]}$ by Problem 1.1(6), while $c_{[-2]} = -2 \neq 2$. So $\iota\circ\psi\circ\varphi = [2]$, and by uniqueness of the dual, $\hat\varphi = \iota\circ\psi$, with $c_{\hat\varphi} = 2$ and $c_\varphi c_{\hat\varphi} = 2 = \deg\varphi$. Finally $\ker\hat\varphi = \ker\psi = \{O_2, (0,0)\} = \varphi(E_1[2](\bar k))$ by (1).

(5) Is there an isogeny $\mathbb{G}_\mathrm{m} \to \mathbb{G}_\mathrm{m}$ of degree 2? If so, does it have a dual?

SOLUTION. **Yes, [2] and [−2]. Their only candidate duals, [±1], have degree 1.** The endomorphisms of $\mathbb{G}_\mathrm{m}$ are the maps $[m](t) = t^m$, $m \in \mathbb{Z}$, and $\deg[m] = |m|$, so the isogenies $\mathbb{G}_\mathrm{m} \to \mathbb{G}_\mathrm{m}$ of degree 2 are [2] and [−2], both with kernel $\mu_2(\bar k) = \{\pm 1\}$. In the sense of the defining identity $\hat\alpha\circ\alpha = [\deg\alpha]$, each has a unique dual: $[n]\circ[\pm 2] = [2]$ means $\pm 2n = 2$, so $n = \pm 1$. And in general the only $[n]$ with $[n]\circ[m] = [\deg[m]]$ is $[\operatorname{sign} m]$, which also satisfies

$[m] \circ [\operatorname{sign} m] = [\deg[m]]$ and is compatible with composition. But for $|m| \geqslant 2$ it fails the other properties of the dual isogeny [Sil09, Theorem III.6.2]: its degree is 1, not $\deg[m]$; the dual of the dual is $[\operatorname{sign} m]$, not $[m]$; and $[m] \mapsto [\operatorname{sign} m]$ is not additive, since $\operatorname{sign}(1+1) \neq \operatorname{sign} 1 + \operatorname{sign} 1$.
Remark. All of this fails because $\deg[m] = |m|$, not m^2.

3. The Tate module

We first describe the Tate module of $\mathbb{G}_{\mathrm{m}}$, the familiar case, and then repeat each step for E. What links the two is the Weil pairing, which you may use without proof.

Proposition 3.1 (The Weil pairing [Sil09, Proposition III.8.1])**.** *Let $m \geqslant 1$ with $\operatorname{char} k \nmid m$. There is a pairing*

$$e_m \colon E[m](k^{\mathrm{sep}}) \times E[m](k^{\mathrm{sep}}) \to \mu_m(k^{\mathrm{sep}})$$

that is bilinear, alternating ($e_m(P,P) = 1$), non-degenerate, and Galois equivariant: $\sigma\big(e_m(P,Q)\big) = e_m(\sigma P, \sigma Q)$ for every $\sigma \in \mathrm{Gal}_k$. Here $E[m](k^{\mathrm{sep}}) = E[m](\overline{k})$, since $\operatorname{char} k \nmid m$. It follows that if (P,Q) is a basis of $E[m](k^{\mathrm{sep}})$ over $\mathbb{Z}/m\mathbb{Z}$, then $e_m(P,Q)$ is a primitive m-th root of unity, and if σ acts on this basis by the matrix $A \in \mathrm{GL}_2(\mathbb{Z}/m\mathbb{Z})$, then $\sigma\big(e_m(P,Q)\big) = e_m(P,Q)^{\det A}$.

Problem 3.2 (The 2-power torsion of $y^2 = x^3 - x$)**.**
Regard E as a curve over $\mathbb{Q}$, fix $\overline{\mathbb{Q}} \subset \mathbb{C}$, and recall that $E[m](\overline{\mathbb{Q}})$ is the kernel of $[m]$ on $E(\overline{\mathbb{Q}})$. Let σ be complex conjugation, acting on $\overline{\mathbb{Q}}$ and on $E(\overline{\mathbb{Q}})$ coordinatewise.

(1) Recall that $T_2(\mathbb{G}_{\mathrm{m}}) \coloneqq \varprojlim_n \mu_{2^n}(\overline{\mathbb{Q}})$, with transition maps $\zeta \mapsto \zeta^2$. Find a $\mathbb{Z}_2$-basis of $T_2(\mathbb{G}_{\mathrm{m}})$, and write down its first three terms. How does σ act on $\mu_4(\overline{\mathbb{Q}})$, and on $T_2(\mathbb{G}_{\mathrm{m}})$?

Solution. **$(-1, i, \zeta_8, \ldots)$. Also σ is $i \mapsto i^3$ on $\mu_4(\overline{\mathbb{Q}})$ and -1 on $T_2(\mathbb{G}_{\mathrm{m}})$.** The map $\zeta \mapsto \zeta^2$ sends the primitive 2^{n+1}-th roots of unity onto the primitive 2^n-th ones, so there are compatible sequences $(\zeta_{2^n})_n$ of primitive roots with $\zeta_{2^{n+1}}^2 = \zeta_{2^n}$. Since ζ_{2^n} generates the cyclic group $\mu_{2^n}(\overline{\mathbb{Q}})$, the maps $\mathbb{Z}/2^n\mathbb{Z} \to \mu_{2^n}(\overline{\mathbb{Q}})$, $a \mapsto \zeta_{2^n}^a$, are isomorphisms compatible with reduction on the left and squaring on the right, and in the limit $\mathbb{Z}_2 \xrightarrow{\sim} T_2(\mathbb{G}_{\mathrm{m}})$. So $(\zeta_{2^n})_n$ is a $\mathbb{Z}_2$-basis, for instance $(-1, i, \zeta_8, \ldots)$ with $\zeta_8 = (1+i)/\sqrt{2} = e^{2\pi i/8}$. Complex conjugation inverts every root of unity: on $\mu_4(\overline{\mathbb{Q}})$ it is $i \mapsto -i = i^3$, and on $T_2(\mathbb{G}_{\mathrm{m}})$ it is multiplication by $-1 \in \mathbb{Z}_2^\times$.

(2) List $E[2](\overline{\mathbb{Q}})$. Using Problem 1.1(6), find the x-coordinates of all $P \in E(\overline{\mathbb{Q}})$ with $[2]P \in E[2](\overline{\mathbb{Q}}) \smallsetminus \{O\}$, and conclude that $\#E[4](\overline{\mathbb{Q}}) = 16$.

Solution. **$E[2](\overline{\mathbb{Q}}) = \{O, (0,0), (\pm 1, 0)\}$. The x_P are $\pm i$, $1 \pm \sqrt{2}$, $-1 \pm \sqrt{2}$.** A point $P \in E(\overline{\mathbb{Q}})$ with $[2]P \in E[2](\overline{\mathbb{Q}}) \smallsetminus \{O\}$ is not in $E[2](\overline{\mathbb{Q}})$, so $4(x_P^3 - x_P) = 4y_P^2 \neq 0$. For such P, evaluating Problem 1.1(6) at P shows that $x([2]P) = ([2]^* x)(P) = e$, that is $[2]P = (e, 0)$, exactly when $g_e(x_P) = 0$: so $x_P = \pm i$ for $e = 0$, $x_P = 1 \pm \sqrt{2}$ for $e = 1$ and $x_P = -1 \pm \sqrt{2}$ for $e = -1$. None of these is a root of $x^3 - x$, so each gives two points $(x_P, \pm y_P) \in E(\overline{\mathbb{Q}})$ outside $E[2](\overline{\mathbb{Q}})$, twelve in all. With $E[2](\overline{\mathbb{Q}})$ this makes 16 points of $E(\overline{\mathbb{Q}})$. They are all of $E[4](\overline{\mathbb{Q}})$, since a point of $E[4](\overline{\mathbb{Q}})$ not in $E[2](\overline{\mathbb{Q}})$ doubles into $E[2](\overline{\mathbb{Q}}) \smallsetminus \{O\}$.

(3) Show that $P_2 \coloneqq (i, 1-i)$ and $Q_2 \coloneqq (1+\sqrt{2}, 2+\sqrt{2})$ are points of $E(\overline{\mathbb{Q}})$, with $[2]P_2 = (0,0)$ and $[2]Q_2 = (1,0)$. Show that (P_2, Q_2) is a basis of $E[4](\overline{\mathbb{Q}})$ over $\mathbb{Z}/4\mathbb{Z}$, and that $E[4](\overline{\mathbb{Q}}) \subseteq E(\mathbb{Q}(\zeta_8))$. Compare with $\mu_8(\overline{\mathbb{Q}})$ and Proposition 3.1.

Solution. For P_2, $x_{P_2}^3 - x_{P_2} = i^3 - i = -2i = (1-i)^2 = y_{P_2}^2$. For Q_2, $x_{Q_2}^2 = (1+\sqrt{2})^2 = 3+2\sqrt{2}$, so

$$x_{Q_2}^3 - x_{Q_2} = x_{Q_2}(x_{Q_2}^2 - 1) = 2x_{Q_2}^2 = 6 + 4\sqrt{2} = (2+\sqrt{2})^2 = y_{Q_2}^2.$$

By (2), $x([2]P_2) = 0$ and $x([2]Q_2) = 1$, and a point of $E(\overline{\mathbb{Q}})$ with x-coordinate 0 or 1 has y-coordinate 0. If $[a]P_2 + [b]Q_2 = O$, doubling gives $[a](0,0) + [b](1,0) = O$, so $a = 2a'$ and $b = 2b'$ are even. Then $[a'](0,0) + [b'](1,0) = O$ makes a', b' even too. So $(\mathbb{Z}/4\mathbb{Z})^2 \to E[4](\overline{\mathbb{Q}})$, $(a,b) \mapsto [a]P_2 + [b]Q_2$, is injective, hence bijective. Every point of $E[4](\overline{\mathbb{Q}})$ thus lies in $E(\mathbb{Q}(i,\sqrt{2}))$, and $\mathbb{Q}(i,\sqrt{2}) = \mathbb{Q}(\zeta_8)$, since $\zeta_8 = (1+i)/\sqrt{2}$. This is exactly $\mathbb{Q}(\mu_8(\overline{\mathbb{Q}}))$. Writing $\mathbb{Q}(E[4](\overline{\mathbb{Q}}))$ for the field generated by the coordinates of the points of $E[4](\overline{\mathbb{Q}})$, Proposition 3.1 forces $\mathbb{Q}(E[4](\overline{\mathbb{Q}})) \supseteq \mathbb{Q}(\mu_4(\overline{\mathbb{Q}}))$: the values of e_4 lie in $\mathbb{Q}(E[4](\overline{\mathbb{Q}}))$ and include a primitive fourth root of unity.

(4) Show that σ restricts to a group automorphism of $E[4](\overline{\mathbb{Q}})$. Compute $P_2 + (1,0)$, and write $\sigma(P_2)$ and $\sigma(Q_2)$ in the basis (P_2, Q_2). Compare the determinant of the resulting matrix in $\mathrm{GL}_2(\mathbb{Z}/4\mathbb{Z})$ with the action of σ on $\mu_4(\overline{\mathbb{Q}})$ found in (1), and explain the agreement using Proposition 3.1.

Solution. **$\sigma(P_2) = [3]P_2 + [2]Q_2$, $\sigma(Q_2) = Q_2$, determinant $3 \equiv -1$.** The group law on $E(\overline{\mathbb{Q}})$ is given by rational functions with coefficients in $\mathbb{Q}$, so $\sigma(P+Q) = \sigma(P) + \sigma(Q)$. In particular $[4]\sigma(P) = \sigma([4]P)$, and σ maps $E[4](\overline{\mathbb{Q}})$ to itself, with inverse σ. Let $R := P_2 + (1,0)$. The line through $(i, 1-i)$ and $(1,0)$ has slope $\lambda = -1$, so

$$x_R = \lambda^2 - i - 1 = -i, \qquad y_R = -\big((1-i) + \lambda(x_R - i)\big) = -1 - i,$$

that is, $R = (-i, -1-i)$. Hence $\sigma(P_2) = (-i, 1+i) = -R = -P_2 - [2]Q_2$. Since P_2 and Q_2 have order 4, this is $[3]P_2 + [2]Q_2$. And $\sigma(Q_2) = Q_2$, since its coordinates are real. In the basis (P_2, Q_2), $\sigma|_{E[4](\overline{\mathbb{Q}})}$ has matrix

$$\begin{pmatrix} 3 & 0 \\ 2 & 1 \end{pmatrix}, \qquad \det = 3 \equiv -1 \pmod 4,$$

and by (1), σ acts on $\mu_4(\overline{\mathbb{Q}})$ by the same exponent 3. This is no accident: by Proposition 3.1, $e_4(P_2, Q_2)$ is a primitive fourth root of unity and $\sigma\big(e_4(P_2,Q_2)\big) = e_4(P_2,Q_2)^{\det} = e_4(P_2,Q_2)^3$.

(5) Let $(P_n)_n$ and $(Q_n)_n$ be compatible sequences, $[2]P_{n+1} = P_n$ and $[2]Q_{n+1} = Q_n$, starting with $P_1 = (0,0)$, $Q_1 = (1,0)$ and P_2, Q_2 as above. Explain why they form a $\mathbb{Z}_2$-basis of $T_2(E)$, and compare with (1).

Solution. The map $(\mathbb{Z}/2^n\mathbb{Z})^2 \to E[2^n](\overline{\mathbb{Q}})$, $(a,b) \mapsto [a]P_n + [b]Q_n$, is an isomorphism for every n. It is injective by the argument of (3), by induction on n: if $[a]P_n + [b]Q_n = O$, doubling gives $[a]P_{n-1} + [b]Q_{n-1} = O$, so $a = 2^{n-1}a'$ and $b = 2^{n-1}b'$ by induction. Then $[a']P_1 + [b']Q_1 = O$ makes a', b' even. It is surjective because $\#E[2^n](\overline{\mathbb{Q}}) = 4^n$, as $E[2^n](\overline{\mathbb{Q}}) \cong (\mathbb{Z}/2^n\mathbb{Z})^2$ [Sil09, Corollary III.6.4]. For $n = 2$ this is the count of (2). These isomorphisms are compatible with reduction on the left and $[2]$ on the right. Passing to the limit, $\mathbb{Z}_2^2 \xrightarrow{\sim} T_2(E)$. This is (1) in rank two: $\mu_{2^n}(\overline{\mathbb{Q}})$ is free of rank one over $\mathbb{Z}/2^n\mathbb{Z}$, while $E[2^n](\overline{\mathbb{Q}})$ is free of rank two.

4. Magma examples

Example 4.1 (The isogenies of Problem 2.2)**.** We check in the function field of E_1 that φ and ψ land where they should, that $\iota \circ \psi \circ \varphi = [2]$, and compare with `Magma`'s own 2-isogeny.

```
E := EllipticCurve([-1,0]);                    // y^2 = x^3 - x
Kx<x> := FunctionField(Rationals());
R<t> := PolynomialRing(Kx);
F<y> := ext<Kx | t^2 - (x^3 - x)>;                          // x = x_1, y = y_1
x2 := F!((x^2 - 1)/x);  y2 := y*(x^2 + 1)/x^2;              // phi^*x_2, phi^*y_2
y2^2 eq x2^3 + 4*x2;                                        // true
x3 := y2^2/x2^2;  y3 := y2*(x2^2 - 4)/x2^2;                 // (psi o phi)^*x_3, ^*y_3
y3^2 eq x3^3 - 16*x3;                                       // true
lam := (3*x^2 - 1)/(2*y);  xd := lam^2 - 2*x;  yd := -(y + lam*(xd - x));
x3/4 eq xd and y3/8 eq yd;                                  // true: [2]
phi := TwoIsogeny(E![0,0]);
aInvariants(Codomain(phi));                                 // [ 0, 0, 0, 4, 0 ]
phi(E![1,0]);                                               // (0 : 0 : 1)
DualIsogeny(phi);
// ... taking (x : y : 1) to ((1/4*x^2 + 1) / x : (1/8*x^2*y - 1/2*y) / x^2 : 1)
```

Magma's dual is $\iota \circ \psi$ written out in the coordinates x_2, y_2 of E_2: $\big((x_2^2+4)/(4x_2),\ y_2(x_2^2-4)/(8x_2^2)\big)$.

Example 4.2 (Division polynomials and $E[4](\overline{\mathbb{Q}})$)**.** For $m \geqslant 2$, `DivisionPolynomial(E,m)` vanishes exactly at the x-coordinates of $E[m](\overline{\mathbb{Q}}) \smallsetminus \{O\}$. Its degree is $(m^2-1)/2$ for odd m and $(m^2+2)/2$ for even m, and its factors for $m = 4$ are x, $x \pm 1$ and the g_e of Problem 1.1(6).

```
[ Degree(DivisionPolynomial(E,m)) : m in [2..8] ];
// [ 3, 4, 9, 12, 19, 24, 33 ]
Factorization(DivisionPolynomial(E,4));
// x - 1, x, x + 1, x^2 - 2*x - 1, x^2 + 1, x^2 + 2*x - 1
Q8<z> := CyclotomicField(8);  i := z^2;  s := z + z^7;  // s^2 = 2
E8 := BaseChange(E, Q8);
P2 := E8![i, 1-i];  Q2 := E8![1+s, 2+s];
2*P2, 2*Q2;                          // (0 : 0 : 1) (1 : 0 : 1)
#DivisionPoints(E8!0, 4);            // 16
E8![-i, 1+i] eq 3*P2 + 2*Q2;         // true: complex conjugation
```

Example 4.3 (Characteristic p)**.** Reducing modulo p, the p-division polynomial collapses, because $\#E[p](\overline{\mathbb{F}}_p) \in \{1, p\}$.

```
for p in [3,5,7,13] do
  Ep  := ChangeRing(E, GF(p));
  psi := DivisionPolynomial(Ep, p);
  print p, IsSupersingular(Ep), Degree(psi), Factorization(psi);
end for;
//  3 true   0  []
//  5 false 10  [ <x^2 + 2, 5> ]
//  7 true   0  []
// 13 false 78  [ <x^6 + 11*x^4 + 5*x^2 + 11, 13> ]
```

For $p = 3, 7$ the curve is supersingular and $E[p](\overline{\mathbb{F}}_p) = 0$. In general, $y^2 = x^3 - x$ is supersingular exactly when $p \equiv 3 \pmod 4$. For $p = 5, 13$ the division polynomial is the p-th power of a polynomial of degree $(p-1)/2$, whose roots are the x-coordinates of the $p-1$ non-zero points of $E[p](\overline{\mathbb{F}}_p) \cong \mathbb{Z}/p\mathbb{Z}$. The exponent p is the inseparable degree of $[p]$. The same computation for $\mathbb{G}_{\mathrm{m}}$ is $t^p - 1 = (t-1)^p$.

Looking ahead. The Weil pairing of Proposition 3.1 also gives $\det \varphi_\ell = \deg \varphi$ for the map φ_ℓ induced by an isogeny on Tate modules ([Sil09, Proposition III.8.6]). Formal groups make c_φ an honest derivative. And Frobenius factors $[p]$ in characteristic p.

References

[Sil09] Joseph H. Silverman, *The arithmetic of elliptic curves*, Second, Graduate Texts in Mathematics, vol. 106, Springer, Dordrecht, 2009. MR2514094 ↑